SECTION 7 · SUBSECTION 7.2
TEST AND VERIFICATION PROGRAM

7.2.1 INTRODUCTION.

The suite is broad, offline and fast: 476 cases in roughly twenty seconds. It is strongest on pure transformations and HTTP contracts, and weakest on the seams between stages. That shape is typical of a system built stage by stage: each agent has detailed unit tests, while the orchestration that connects them is covered only indirectly. Read a green run as evidence that the parts behave, not that the whole path has been exercised (figure 7.2-1).

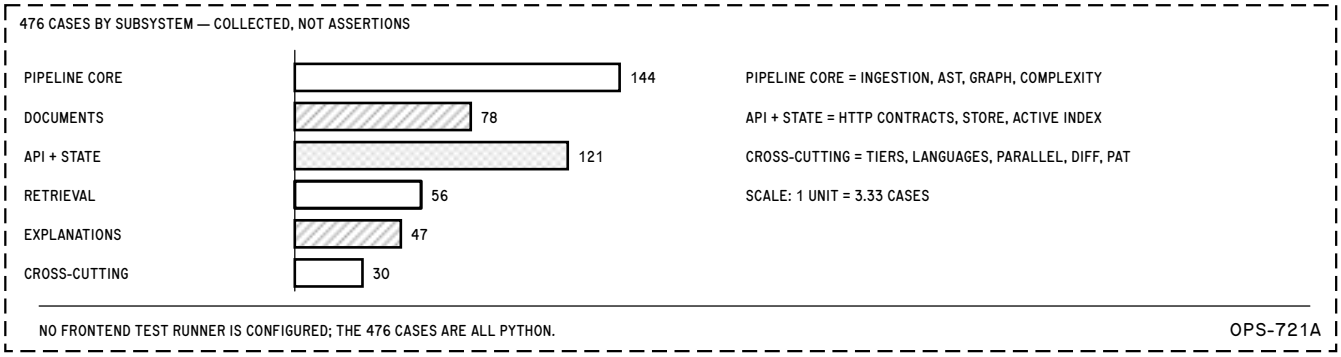


Figure 7.2-1. Case Distribution Across Subsystems

7.2.2 FUNCTIONAL DESCRIPTION.

Unit tests build source strings, parse them and assert on dataclasses; they do not need a repository. API tests import the FastAPI application, replace the pipeline submission with a fake, and drive routes through *TestClient*. Store tests run the same lifecycle twice, once against the memory backend and once against *fakeredis* with WATCH semantics. The active-index tests simulate contention by mutating keys between a read and a write, and cover lost sentinels, external members and interrupted reconciliation.

The resilience tests are the most informative for an operator: partial model failure, missing collection, missing key, terminal-record protection and retry exhaustion all have explicit cases. Multi-language tests cover symbol extraction and complexity for Go, Rust, Java and C++, plus alias and layout resolution. Two cases assert that concurrent parsing and scoring return the same state as sequential execution.

Table 7.2-1. Test Layers and What They Establish

Layer	Cases	What a pass establishes
Ingestion and filters	22	URL forms, extension and directory rules, size ceiling, language labels
Parsing and symbols	34	Python and JS declarations, docstrings, imports and parse-error capture
Graph resolution	38	Internal versus external imports, cycles, order, stats, PageRank fallback
Complexity and risk	50	Branch counts, threshold boundaries, precedence, coupling from graph stats
Retrieval	56	Chunk identity, collection naming, metadata filters, batch embedding
Explanations	47	Selection priority, context budget, prompt content, sanitisation, partial failure
Documents	78	Section presence, order, tables, empty-state safety, heading hierarchy
HTTP and state	121	Status codes, validation, lifecycle, health, duplicate submission, index drift
Cross-cutting	30	Tiers, four extra languages, parallel equivalence, diff, private-repo headers

7.2.3 PERFORMANCE AND DESIGN DATA.

Table 7.2-2. Observed Run and Uncovered Paths

Item	Value
Result	476 passed
Wall time	20.40 s
Files	15
Failures / skips	0 / 0
Network calls	none observed
Frontend tests	none configured
Worker loop	handler only
CLI savers	not end-to-end
Incremental delete path	not covered
Real Chroma persistence	not covered

WARNING

A green suite does not cover the undefined *logger* in incremental chunk deletion, the missing on-disk *agent_context.md*, the method-count defect in the agent context, or the cleared *raw_contents*. Each sits on a seam between tested units.

TEST CHECK	ENGINEER
1 RUN - full suite, record time	
2 READ - first failure, not the count	
√ KNOW - what is not tested	
3 ADD - a seam test before a fix	

OPS